

12 (long)

(a) Sketch a plot of cyclic stress amplitude against fatigue life, indicating the endurance limit, and the regimes of low- and high-cycle fatigue. Explain what is meant by the endurance limit. Briefly describe the main difference between low cycle fatigue and high cycle fatigue. For each mechanism, explain how the number of cycles to failure N_f can be modelled. [8]

(b) Table 2 shows high cycle fatigue data for an aluminium alloy, from a series of tests in which the stress had a peak-to-peak variation $\Delta\sigma$ about a mean stress of zero.

$\Delta\sigma$ (MPa)	200	300	400
N_f	6.15×10^8	5.06×10^6	1.68×10^5

Table 2

Use a suitable plot of the data to confirm that Basquin's law is applicable. Estimate the number of cycles to failure if a specimen of the same alloy is subjected to a fatigue test where the stress varies from -175 MPa to 175 MPa. [6]

(c) Using sketches and equations as appropriate, explain how you would predict the fatigue life in the following situations:

- (i) loading cycles at constant stress range with a constant non-zero mean stress;
- (ii) loading cycles of varying stress range about a zero mean stress. [6]

(d) Bicycle spokes are manufactured from the aluminium alloy in part (b). The alloy has an ultimate tensile strength of 600 MPa. In a completed wheel, the spokes are pre-tensioned to a stress of 300 MPa. The training regime of a mountain bike rider subjects a typical spoke in the rear wheel to a number of loading regimes, with maximum and minimum stresses σ_{max} and σ_{min} . The proportion of stress cycles spent in each of four loading regimes are shown in Table 3.

Given that the outside diameter of a mountain bike tyre is 640 mm, and the rider trains for 120 km per day, calculate the fatigue life in days of the spokes in the rear wheel. [10]

σ_{max} (MPa)	σ_{min} (MPa)	Proportion of stress cycles (%)
300	250	80
300	200	12
300	150	6
300	100	2

Table 3

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Paper 2

Numerical Answers

Section A (Structures)

1. $2WL/AE$ to right
2. $P(r_o-r_i)/4t$ and $P(r_o-r_i)/2t$
3. a) Vertical: 22.5 kN, 47.5 kN, Horizontal 1125 kN
(b) 0.38m
4. (a) $M = wL^2/4$
(b) $x = L/\sqrt{3}$
5. (a) $8SL^3/3EI$ downwards
b) $(8S-W)L^3/12EI$ downwards
c) $WL^3/15EI$ upwards
6. a) 270 kN

Section B (Materials)

7. (b) $P_s = \exp \left[- \left(\frac{\alpha}{R^4 \sigma_o} \right)^m \left(\frac{2\pi L}{V_o} \right) \left(\frac{R^{m+2}}{m+2} \right) \right]$, (c) $P_s = 0.96$
9. (b) $E_{\text{comp}} \sim 0.262 E_f$, (c) ~ 31 GPa
10. (a) $\sigma_o = 60$ MPa, (b) $n \sim 1/2$
11. (c) biocomposite, ~ 234 MJ
12. (b) $N_f \sim 8.1 \times 10^5$ cycles
(d) life = 3471 days